PV101 Day 1: First Day Orientation

Student Packet | CTS Cohort 6 | 2026-07-06

Name: _____ Group: _____

Today in One Sentence

Main idea

Solar work starts with learning the parts, asking good questions, and verifying what is actually true.

Today You Will Be Able To

- Explain what PV means.
- Name the main parts of a small solar system.
- Describe why we verify instead of guessing.
- Work in a group to map a simple RV solar system.
- Ask a useful question about the Cohort 6 RV solar capstone.

Key Words

Word	Meaning
PV	Photovoltaic. A way to turn light into electricity.
Solar module	The panel students can see and touch. It makes DC electricity when light hits it.
Array	A group of solar modules working together.
DC	Direct current. Battery and solar side of many small systems.
AC	Alternating current. Common wall-outlet power.
Charge controller	Controls charging between the solar module and battery.
Battery	Stores energy for later use. Cohort 6 default battery type: 12.8V LiFePO4.
Fuse	A protection device that opens when too much current flows.
Load	Anything using power, such as lights, fan, pump, or USB charger.
Inverter	Changes DC battery power into AC power for AC loads.
Verify	Check what is actually true before trusting memory, labels, or guesses.

Simple Solar System Path

Copy this path

Sun -> Solar Module -> Charge Controller -> Battery -> Fuse Panel -> Loads

Battery -> Inverter -> AC Loads

Group Activity 1: RV Loads

In your group, list things in an RV that might need power. Then mark each one as DC, AC, or unknown.

#	RV Load	DC / AC / Unknown
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		

Group Activity 2: First System Map

Draw a simple map showing how energy moves from the sun to something that uses power.

_____

_____

_____

_____

_____

_____

_____

_____

Labels vs Verification

Finish the sentences.

A label is helpful because:

_____

_____

A label is not enough because:

_____

_____

One thing I can verify before trusting a system is:

_____

_____

Exit Ticket

1. One thing I understand from today:

_____

_____

2. One thing I am still confused about:

_____

_____

3. One thing I want to learn or try in PV101:

_____

_____